UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering

Universiti Sains Malaysia, Engineering Campus

SEMESTER 1, 2026/2027

EEM 441

Control & Instrumentation Laboratory

Experiment 7

DC MOTOR SPEED MEASUREMENT

Date : ______________________________

Group No. : ______________________________

Group Members : ______________________________ Matrix # __________

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Experiment 7

Speed Motor Measurement Using Raspberry Pi

Objective

Design and implement a speed measurement system using a Raspberry Pi. The system should accurately measure the speed of a moving object using sensors, process the data, and display the results. This lab encourages students to apply principles of embedded systems, sensor interfacing, and signal processing.

Learning Outcomes

By the end of this lab, students will:

1. Understand the basics of speed measurement and its applications.
2. Develop skills in interfacing sensors with a Raspberry Pi.
3. Implement a Python/C/C++ program to capture, process, and display speed data.
4. Gain experience in open-ended problem-solving and system design.

Introduction

In this open-ended lab, students will design and implement a system to measure the rotational speed of a motor using a Raspberry Pi. This project challenges students to develop and build the necessary circuitry to process signals from a sensor trainer and interface it effectively with the Raspberry Pi. The goal is to create a complete system that captures the motor's rotational data, processes the information, and displays the measured speed. Through this hands-on experience, students will deepen their understanding of embedded systems, sensor integration, and data visualisation while developing practical problem-solving skills. An overall block diagram for the measurement system is shown in Figure 7.1.

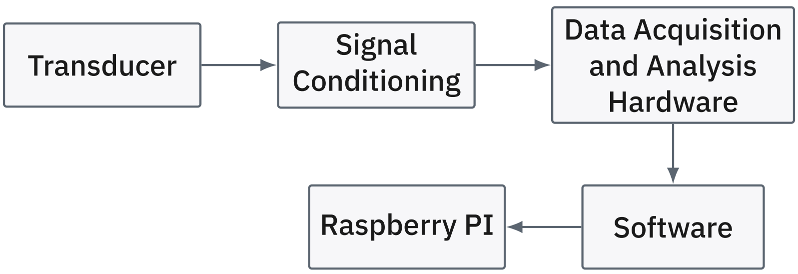

Figure 7.1: Data Acquisition System.

Equipment / Software Required

- Raspberry Pi (Raspberry Pi 4B)
- Python / C/C++ installed on the Raspberry Pi
- HPAC-08 trainer

Procedure

1. **Raspberry Pi setup.** You may set up with a Windows-based system or Linux-based system. Please check the resources in the References section. Software: you can use either Python or C/C++ with Microsoft Visual Studio Code.
2. Design an appropriate circuitry to convert the signal from the sensor to a signal suitable for the Raspberry Pi, including signal conditioning and signal conversion. Show all the necessary calculations for the measurement of speed.
3. **Displaying results.** Print the calculated speed to the terminal or display it on a small screen connected to the Raspberry Pi.
4. **Testing and verification.** Compare the results measured with the Raspberry Pi and the signal from the sensor using a DSO.

Table 7.1: Speed measurement: Raspberry Pi vs. DSO reference.

Input voltage	Velocity (rad/s)(Raspberry Pi)	Velocity (rad/s)(DSO)

Deliverables

1. **Code:** a programming script that accurately measures and displays speed based on sensor inputs.
2. **System setup diagram:** a diagram showing the placement and wiring of sensors to the Raspberry Pi.
3. **Documentation:** a report that explains the setup, the speed measurement process, sample readings, discussion, and any challenges encountered.
4. **Demo:** a short presentation demonstrating the working system and the collected data.

Evaluation Criteria

- **Accuracy of measurements:** consistency and accuracy of speed readings.
- **Code functionality:** proper functioning of the program without errors.

- **Creativity in design:** unique solutions for sensor placement, code optimisation, or data handling.
- **Clarity of documentation:** clear and thorough explanation of the approach, results, and challenges.

Conclusion

This open-ended lab provides hands-on experience in designing a basic speed measurement system, enhancing students' understanding of sensors, timing, and embedded programming. It also encourages troubleshooting and creativity in problem-solving, as there are multiple solutions for achieving accurate speed measurement.

References

- Setting up Raspberry Pi for the first time: https://www.tomshardware.com/how-to/set-up-raspberry-pi
- https://core-electronics.com.au/guides/temperature-sensing-with-raspberry_pi/
- https://www.cl.cam.ac.uk/projects/raspberrypi/tutorials/temperature/
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- https://code.visualstudio.com/docs/setup/raspberry-pi
- Coding on Raspberry Pi remotely with Visual Studio Code: https://www.raspberrypi.com/news/coding-on-raspberry-pi-remotely-with-visual-studio-code/
- https://besomi.com/coding-on-raspberry-pi-remotely-with-visual-studio-code/
- https://github.com/gloveboxes/Raspberry-Pi-with-Visual-Studio-Code-Remote-SSH-and-C-or-CDevelopment